

Representation Convergence — Results Summary

Every experiment's measured outcome in one place. Thesis: independently trained models share representations up to a linear transformation. All figures are held-out; controls and chance rates are quoted alongside. Full method and caveats in [Final_Project_Report.pdf](#) (69 pp).

The findings at a glance

#	finding	headline evidence
1	Convergence is linear	MLP = ridge to 4 decimals (0.9183 = 0.9183), reproduced on two independent runs of a fresh corpus
2	Related, not rigid	Procrustes R2 0.038 vs ridge 0.592; 69x spread in the map's singular values
3	Deployable	95% of ceiling at 0.79 MB; verification AUC 0.999
4	Cross-modal without joint training	DINOv2 -> bge 40.9-58.3% of a matched ceiling, up to 358x chance
5	Alignment scales with capacity	+15.3 points of ceiling per decade of parameters, six monotone measures
6	Isotropy, not similarity-training	whitening a raw LM space reaches 102.6% / 101.4% of a contrastive encoder at two scales
7	Correspondence is self-identifying	Gromov-Wasserstein 39.7% exact = 595x chance, zero paired examples
8	One shared space, used	hub transfer 92.9-96.5% of native on encoders never seen, incl. cross-lineage
9	The mechanism predicts	pre-registered encoder ladder, rank correlation +1.00

Meta-finding, measured twice, corrected from three. Isotropy helps when READING a correspondence (C.11) and hurts when FINDING one (C.12) - one geometric property, opposite prescriptions. The width cliff was previously counted as a third instance; it is not - the first two are about isotropy, the cliff is about rank (E.8).

Experiment A — scientific proof (GPT-2 vs Pythia-160M)

metric	fail	pass	measured
CKA diagonal mean (trained)	<0.15	>0.35	0.424
CKA mean, random control	>0.20	<0.15	0.067
trained / random ratio	<2x	>=3x	6.3x
stitching R2, peak layer (L11)	<0.50	>0.70	0.797
stitching R2, random control	>0.40	<0.30	0.173
depth-slope signature	both flat	opposite	0.59->0.80 vs 0.38->0.09

Verdict: PASS. The trained curve rises with depth while the random control decays - opposite slopes, an 8x gap at L11. Run 1 (1,646 samples) failed on sample starvation; run 2 (10,000) passed decisively, identical pipeline.

Experiment B — the deployed adapter (MobileCLIP-S1 -> SigLIP 2)

quantity	value
Ridge adapter: held-out R2 / mean cosine	0.592 / 0.900
Procrustes (rotation only): R2	0.038 (anisotropy, not just orientation)
MLP adapter: best cosine	0.903 (+0.003 -> relationship is linear)
Retrieval R@1 / R@5 / R@10 (ceiling)	0.630 / 0.885 / 0.944
Retrieval with linear adapter	0.557 / 0.843 / 0.918
as % of ceiling	88.4% / 95.3% / 97.2%
raw cross-space baseline	0.001 / 0.005 / 0.010 (exact chance)
verification / ranking / membership	AUC 0.999 / R@1 0.987 / AUC 0.889
hubness N10 skew: raw -> adapted	4.28 -> 1.61 (ceiling 1.41)
deployed artifact	Adapter.mlpackage, 0.79 MB fp16, iOS 16+

Series E — cross-modal, scaling, geometry

experiment	result
E1 capacity sweep (DINOv2 -> bge)	small/base/large R@1 40.9% / 53.5% / 58.3% of ceiling; +15.3 pts per decade, fit R2 0.949
E1.1 swap text encoder to GPT-2	R2 retains 71%, retrieval collapses to 34%; cosine 1.000 while R@1 0.160
E1.2 isotropy rescue	whitening lifts R@1 0.130 -> 0.478 = 102.6% of a 1.2B-pair contrastive encoder; eff. rank 6.5 -> 490.8; iso-retrieval corr +0.94
E1.3 encoder ladder (pre-registered)	P1 rank correlation +1.00 (confirmed); P2 partial - isotropy necessary, not sufficient

Series F — the contrast condition (deliberate training)

experiment	result
F1/F2 capacity test	MLP converges to EXACTLY linear: cosine 0.9183 = 0.9183, gain +0.0000, reproduced on two runs from a fresh 38k-row corpus (74.2 rows/dim)
F5 data test	linear refit on 38k gives R@1 +0.0244 (bootstrap CI excludes 0); 90%-of-ceiling gate reported NOT met at 89.0%
F3 unfreeze bottleneck	built and gated shut - running it would cost the clean 'no pretrained weight modified' claim; the gate is the contribution

Series C — pair-free recovery

quantity	value
Gromov-Wasserstein exact matches	39.7% = 595x chance, zero pairs
map fitted from that assignment	R@1 0.252 (40% of ceiling), verification AUC 0.906
negative 1: ICP refinement	degrades monotonically (converges on its own errors)
negative 2: whitening before matching	39.7% -> 0.1% (destroys matching)
scope (C.12.1)	both encoders contrastive image-text - the WEAKEST-independence pair; result is bounded accordingly

Series G — the shared hub

experiment	result
G1 seven-space hub, transfer	R@1 0.451 at HUB_DIM 512 (AUC 0.986); transfer cliff at 512->768; vs 0.445 at five spaces - UNCHANGED (within noise), which is the useful result
G1 composition (scale-free R@1)	95.6% retention over 36 of 41 pairs; the 5 exceptions all write INTO GPT-2
G1 read/write asymmetry	GPT-2 as source 95.1%, as target 24.8% - a collapsed space is readable out of, not writable into; tracks effective rank across bge/BERT/GPT-2
G3 compatibility (vs bespoke hubs)	42 of 42 ordered pairs SHARED, mean retention 103.6%; control at chance (0.004 vs 0.001)
G4 cross-lineage transfer	SigLIP 2, an INDEPENDENT lineage, retains 94.2% of a natively-fitted head - inside the 93.8-96.5% within-family band; control at chance. Cross-lineage, not simultaneously cross-objective; still a ViT
G5 scale of agreement	CKNNA: 21/21 pairs higher locally than at the global limit; agreement is LOCAL. The raw ratio-to-chance curve is NOT the evidence - it is bounded by N/k (C.13.9)
G5 what explains it (exploratory)	objective (+53x) > lineage (+44x) > modality (+29x); top pair bge-SBERT shares objective, no lineage
G5 CKNNA (the literature's metric)	21 of 21 pairs score higher locally than at the global limit; rank correlation +0.81 with our own measure. The local-vs-global conclusion does not depend on the instrument
G5 permutation calibration	k-NN null = k/N exactly (already calibrated); CKA null +0.094 and RISES with width. Correcting it WIDENS the objective-over-lineage gap from +0.024 to +0.181 - the confound had favoured the lineage pair
G6 manifold characterisation	intrinsic dimension 11.2-19.9 inside 768-2048 wide spaces (<3% of ambient); Procrustes across 21 pairs mean 0.194 vs ridge 0.667; CCA shows cross-modal pairs share as few as 6 of 64 directions

G7 what the hub does to local structure	on healthy spaces NEUTRAL, and now BOUNDED (K3): cluster-bootstrap 95% CI -0.038 to +0.024, MDE 0.034 vs a pre-registered 0.050 - a demonstrated null, not absence of evidence. Six pairs from four encoders, so the interval resamples encoders. Rebuilt -0.010 vs original +0.010: no stable sign. Degenerate pairs +0.151 (C.11 rescue, 4th instance)
G7 the width cliff	reproduced on ZERO-SHOT transfer: 0.489 at 512 -> 0.312 at 768. Now EXPLAINED and it is a NUMERICAL ARTIFACT, not a finding about representations: least squares divides by a near-null singular value that appears when hub width reaches the source's rank (E.8). Seven accounts falsified first

Series H / V — certification

check	result
H1 artifact battery	47 checks, NO hard failures; 2 flags, both the documented GPT-2 collapse, reproduced independently in two files
rows per input dimension	every fit 5.9-9.8, above the pre-registered floor of 5
alignment (shuffle test)	honest fits beat shuffled by 0.545-0.863 vs a 0.2 threshold
V1 independent reproduction	verification AUC 0.9994, R@1 0.987, 69x knob spread - three reported figures from a separate code path

Series G / K - addenda, 23-24 August 2026

experiment	result
G0-exact protocol reproduced	original construction recovered from G4's notebook (4 spaces, per-space scaling, random seed-0 split, bge targets, alpha 1e-2). SigLIP 94.2% against a published 94.2%; DINOv2-base 96.5% vs 95.9%, -large 93.8% vs 92.9%. LEVELS, not orderings
G4-conv cross-ARCHITECTURE	ConvNeXt-base (convnet, IN-22k) 96.7% of native - ABOVE the 93.8-96.5% band and above SigLIP's 94.2% in the same run. Architecture does not bound the claim. Control 0.001 = chance. Objective still confounded
G8 four cliff statistics	0.55, 0.39, 0.67, 0.88 against transfer's 4.03 on a pre-registered discontinuity standard. ALL FOUR FAIL - each smooth where the outcome is a step
G9/G11/G12 cliff LOCATED	the cliff sits at the SOURCE encoder's ambient dimension. Predicted before measuring for two encoders that could not have suggested it; 6 of 6 curves hit (768, 1536, 2048) under both head targets; three added widths that are nobody's dimension show nothing
G13 a false negative, ours	an ablation and a rank check appeared to falsify the account; both were flawed. Entropy rank (573.7) measures variance concentration, not algebraic rank (768), and the ablation zeroed the TEST encoders rather than the HEAD's fit. Recorded because the error was ours
G14/G15/G16 MECHANISM	CONFIRMED to single-direction resolution. The head's coordinates have algebraic rank d, so at width d the last direction is near-null: sigma 0.0054 vs 10.79 one step earlier (img_small), 7.6e-07 vs 8.73 (img_base). Transfer holds to k=760 then 0.009; holds to k=1528 then 0.001. Truncation (0.878) or head alpha 1e3 (0.917) repairs it. A NUMERICAL ARTIFACT of the fit, not a property of shared coordinates
K1 raw-space Procrustes	ORDERING reproduces with no PCA step anywhere: bge-SBERT +0.738 top, GPT-2 pairs 0.059-0.069 bottom - Table C12k is not a reduction artefact. Similarity transform 0.281 vs ridge 0.667 (42%)
K2 G7 re-run	stratification reproduces: GPT-2 preservation 0.262 (pub 0.257), Spearman +0.891 (pub +0.820), degenerate pairs +0.122 (pub +0.151). SBERT again healthy at 0.135
K3 the null, bounded	cluster-bootstrap 95% CI -0.038 to +0.024, MDE 0.034 against a pre-registered SESOI of 0.050 - a DEMONSTRATED null. Six pairs from four encoders, so encoders are resampled, not pairs
G10 hub composition	Q1: text-MEMBER identity moves transfer 6.1pt at fixed count (95.3% bge-only vs 89.2% bert-only), 11.2pt for presence vs absence (85.5% to 96.7%) - qualifies E.6's low-reach reading. Q2: target-in-hub overlap confirmed at +3.0pt beyond the +6.8pt generic text-member benefit (mean matching +9.8pt vs mean non-matching +6.8pt). Declared, not disqualifying - held out throughout

Nine findings, and eleven explanations tested and recorded as FALSE (two of them the project's own). Eight of the eleven are accounts of the width cliff, which is now EXPLAINED - and explained as a numerical artifact of fitting the head by least squares on rank-deficient coordinates, not as a property of shared representations. A later addition (G10) qualifies E.6 rather than the cliff: hub-composition identity moves transfer independently of reach. Every number here is held-out and quoted against a control. The addenda reproduce the original hub protocol exactly, SigLIP's published 94.2 returning to the decimal, so their transfer figures are levels rather than orderings.

Cached vectors, dimensions and splits

Every measurement in this report runs on cached embeddings, so the substrate is fixed and auditable. Row counts differ because the caches were written over a growing image list; where a space has more rows than the hub uses, the first 9,533 are taken - the same deterministic prefix every cache was written against. Mean row norm is listed because it spans a factor of 234 across spaces, which is why the hub scales each space before concatenating.

space	cache file	rows	dim	row norm	used in
img_small	e1_img_ckpt_dinov2-small_cls+patch	9,533	768	54.70	hub, G1/G4, all pair tests
img_base	e1_img_ckpt_dinov2-base_cls+patch	18,066	1,536	52.91	hub; first 9,533 used
img_large	e1_img_ckpt_dinov2-large_cls+patch	23,755	2,048	50.83	hub; first 9,533 used
txt_bge	crossmodal_pairs.npz [txt]	18,066	1,024	0.89	hub member AND head target
txt_gpt2	crossmodal_pairs_gpt2.npz [txt]	18,066	768	207.78	G1 hub; C.11 collapse case
txt_bert	e13_txt_bert.npz	18,066	768	9.15	G1 hub; encoder ladder
txt_sbert	e13_txt_sbert.npz	18,066	768	2.43	G1 hub; best target (101.0%)
SigLIP 2	e1_img_ckpt_siglip2-base-patch16-224	9,533	768	11.38	HELD OUT - cross-lineage
ConvNeXt	e1_img_ckpt_convnext-base-224-22k	9,533	1,024	16.70	HELD OUT - cross-architecture
Experiment B	pairs.npz: mob_img, sig_img, sig_txt	4,000	512, 768	1.00	adapter, B2-B10, C1
distillation	distill_corpus.npz: sig, mob	39,999	768, 512	1.00	F-series

Splits, pairs and counts

- Hub protocol (G1, G2, G4, G0-exact): 9,533 shared images, 8,533 train / 1,000 eval, random permutation, seed 0.
- Experiment B (adapter): 4,000 COCO images, 3,000 train / 1,000 eval, seed 0. ONE caption per image.
- E and G series: FIVE COCO captions per image, embedded then averaged - not comparable to B's single-caption protocol. Every 'percent of ceiling' is measured against a ceiling re-measured under the same treatment.
- Distillation corpus (F1-F5): 39,999 rows, 37,999 train / 2,000 eval, teacher-supplied targets, no captions.
- Pair counts: 7 spaces give 21 unordered and 42 ordered pairs. The hub serves 4 spaces (G4, the published 94.2%) or 7 (G1).
- Rows per input dimension: every fit between 5.9 and 9.8, above the pre-registered floor of 5 (H1).
- Deployed artifact: 512 x 768 ridge matrix, 0.79 MB as Core ML fp16.